LEARNER GUIDE

Changing the subject

Formulae and Rearranging · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The **subject** of a formula is the letter on its own: in v = u + at the subject is v. **Changing the subject** makes a different letter stand alone. It works exactly like solving an equation: find what was done to the letter you want, then undo it, **in reverse order**, doing the **same to both sides**. Each operation is undone by its inverse: – undoes +, ÷ undoes ×, and a square root undoes squaring.

What you are learning to notice

- The last thing done to the letter is the first thing to undo.
- Do the same to the whole of both sides. Dividing one side by 3 divides every term on it.
- Check with numbers: choose values that make the original formula true, and test the new formula with them.
- A rearranged formula answers the new question for any values, straight away.

Moving through it, one step at a time

Write the formula as a machine: start with the letter you want, and list what is done to it in order. Then undo the steps from the last to the first, doing each one to both sides and writing a new line each time. Finally, check with numbers.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Two steps

Make t the subject of v = u + at.

1. What happens to t? First × a, then + u.	$t \to \times a \to + u \to v$
2. Undo + u first: subtract u from both sides.	$v - u = at$
3. Then undo × a: divide both sides by a.	$(v - u)/a = t$
4. Check with u = 2, a = 3, t = 4: v = 14.	$(14 - 2)/3 = 4$ ✓

t = (v – u)/a.

Example 2 — Rearrange, then use

F = 1.8C + 32 converts °C to °F. Make C the subject and convert 212 °F.

1. Subtract 32 from both sides.	F – 32 = 1.8C
2. Divide both sides by 1.8.	C = (F – 32)/1.8
3. Substitute F = 212.	C = 180/1.8 = 100

C = (F – 32)/1.8, so 212 °F is 100 °C.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (solving one- and two-step equations, and inverse operations), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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